NTFS Permissions Lab

Folder permissions, inheritance, and access testing on Windows

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Lab dates	2 November 2026 (setup and build), 4 November 2026 (final test result)
Computer	HP laptop, host name DESKTOP-VV1G34P, Windows 10
Main account	VOICES Telecoms (administrator)
Test account	Tunde (standard user), created for this lab
Lab folders	C:\Finance and C:\Finance\Payroll
Goal	Show that I understand NTFS permissions and inheritance by building a restricted folder and proving it works from a second user account.

1. Summary

I built a small version of the folder structure from my course diagram: a **Finance** folder with a **Payroll** folder inside it. I then changed who is allowed in each folder and tested the result by signing in as a different user, **Tunde**.

Result: Tunde can open Finance and see that Payroll exists, but when he tries to open Payroll, Windows blocks him with *"You don't currently have permission to access this folder."* This confirms the permissions work as designed.

What this lab shows I can do:

- Read and explain the Security tab and the Advanced Security Settings window.

- Explain permission inheritance, and tell inherited permissions (faded ticks) from explicit ones (solid ticks).

- Disable inheritance on a folder and remove a group from it.

- Spot a risky permission (Everyone with Full control) and explain why it is dangerous.

- Verify access by testing from a second account, not just by reading the settings.

2. Background and preparation

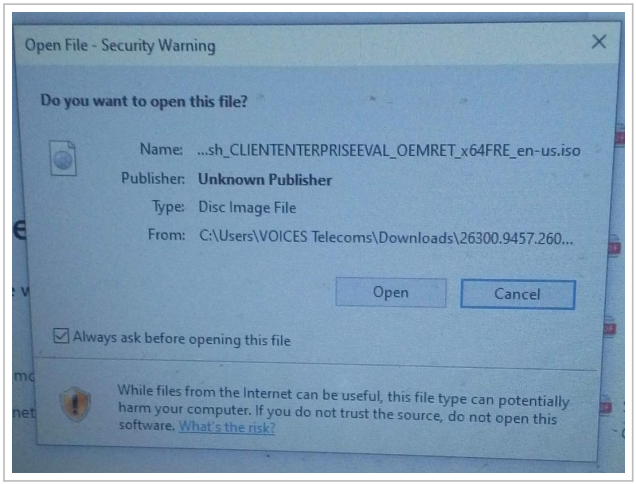
Before the permissions work, I prepared my lab environment.

Step A: Download Windows 11 Enterprise evaluation

I downloaded the Windows 11 Enterprise evaluation ISO from microsoft.com. Windows showed an "Unknown Publisher" warning, which is normal for ISO files from the Internet.

I chose to wait before opening it. I plan to run it in Oracle VirtualBox, the same way I run Ubuntu.

Next I planned the Windows Server 2022 evaluation ISO from the Microsoft Evaluation Center (free for 180 days, activation within 10 days) for the Active Directory lab that follows this one.



Screenshot: Open File - Security Warning for the Windows 11 Enterprise evaluation ISO in my Downloads folder.

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3. Concepts I practiced

NTFS is the file system Windows uses. It lets an administrator decide exactly who can do what to each file and folder.

Permission	What the user can do
Read	View files and folders
Read & execute	Read, and also run programs
Write	Create and change files
Modify	Read, write, and delete
Full control	Everything in Modify, plus change permissions and take ownership

Inheritance. A folder passes its permissions down to everything inside it. In the Security tab, an inherited permission shows as a **faded** tick and a permission set directly on that folder shows as a **solid** tick.

Groups, accounts and membership. An account is one login (for example VOICES Telecoms or Tunde). A group is a bucket of accounts (for example Administrators or Users). Permissions are

best given to groups, not individuals.

Access is a list of names. If neither your account nor a group you belong to is on the list, you have no access.

Share vs NTFS. When a folder is shared over a network, effective access is the stricter of the share permission and the NTFS permission.

The course diagram

My course diagram shows D:\Shares\Finance with Finance-Readers: Read, a Payroll folder that inherits Read plus HR: Modify, and salaries.xlsx that inherits both. The red box on the Security tab marks a mistake: **Everyone** has **Full control**, which lets any user change or delete anything and makes the other settings pointless. The fix is to remove Everyone.



Screenshot: Permission Inheritance diagram from the course, with the Everyone / Full control entry highlighted.

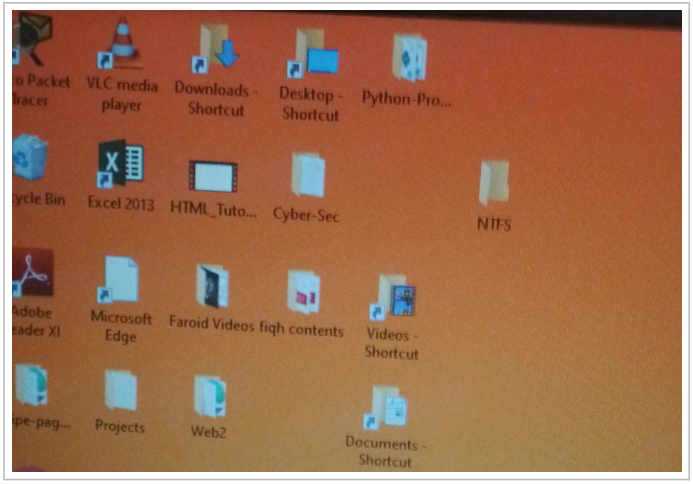
4. The lab, step by step

Each step below shows what I did and the screenshot taken at that moment.

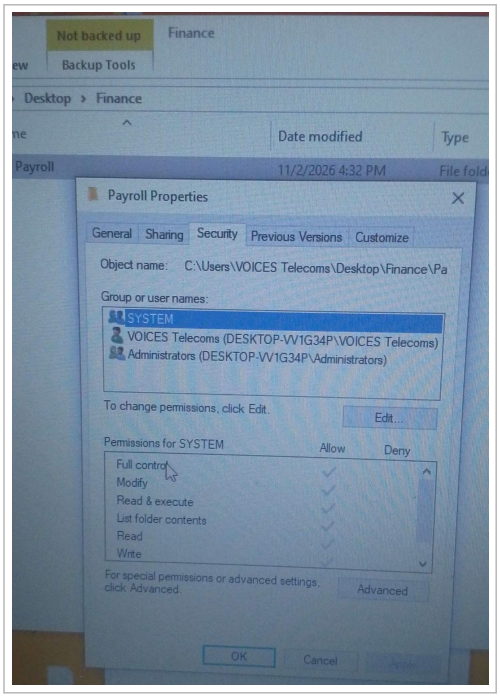
Step 1: Create the practice folders

I created a working folder named NTFS on my Desktop. Inside it I built the structure for the lab: a **Finance** folder, a **Payroll** folder inside Finance, and a text file inside Payroll (created with right-click > New > Text Document).

Files and folders are created from File Explorer with right-click in an empty space, then New.



Screenshot: My Desktop with the NTFS practice folder.

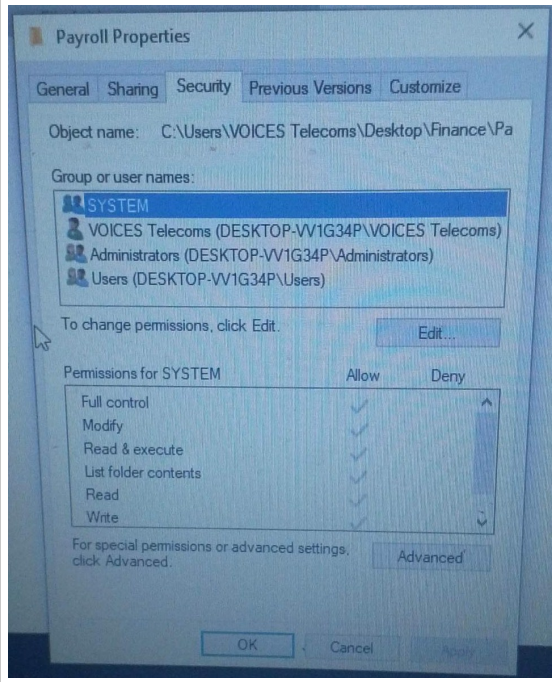


Step 2: Observe inheritance on Payroll

I right-clicked **Payroll**, opened **Properties > Security**, and read the permission list.

Payroll listed **SYSTEM** (Windows itself), **VOICES Telecoms** (my account) and **Administrators** (the admin group). The ticks were **faded**, which means the permissions are inherited from the parent folder.

Screenshot: Payroll Properties, Security tab. Faded ticks show inherited permissions.



Step 3: Prove inheritance by changing the parent

On the **Finance** folder I clicked Edit > Add, typed **Users**, and gave the group Read access.

When I reopened Payroll, **Users** had appeared in its list without my touching Payroll. That is inheritance working: a change on the parent flowed down to the child.

Screenshot: Payroll Properties now lists Users, inherited from Finance.

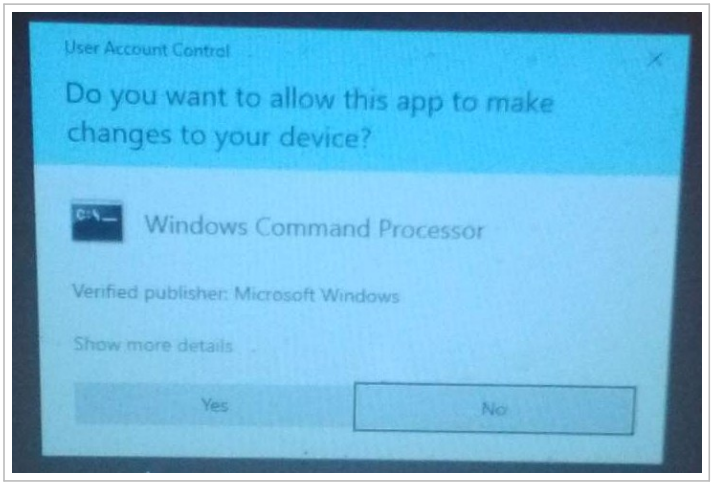
Step 4: Create a test user with Command Prompt

To test properly I needed a second person. I opened Command Prompt as administrator. Windows asked "Do you want to allow this app to make changes to your device?" The publisher was verified as Microsoft Windows, so I clicked Yes.

I then ran the command below, and Windows replied that it completed successfully:

```
net user Tunde <password> /add
```

This created the standard (non-admin) account **Tunde**. The test password is not recorded in this report.

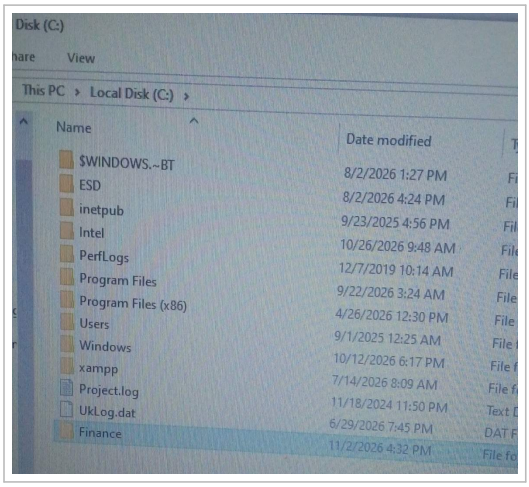


Screenshot: User Account Control prompt for Windows Command Processor (verified publisher: Microsoft Windows).

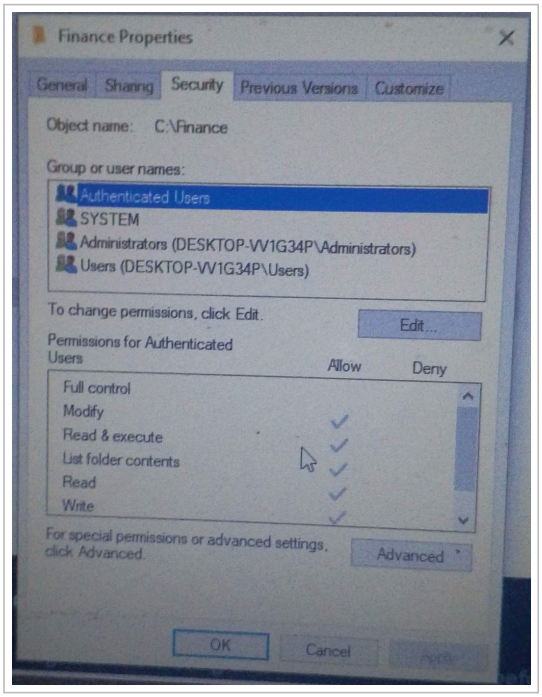
Step 5: Move Finance out of my profile

My Finance folder was inside my own user profile (C:\Users\VOICES Telecoms\Desktop), which Windows keeps private. Other accounts would be blocked from everything there, which would spoil the test.

I cut Finance and pasted it onto the C: drive so it is shared space. It is now at **C:\Finance**.



Screenshot: Local Disk (C:) with the Finance folder at the bottom of the list.



Step 6: Find the risky permission on Finance

On **C:\Finance** I opened Properties > Security. Besides SYSTEM, Administrators and Users, the list contained **Authenticated Users** with **Modify** (inherited from the C: drive).

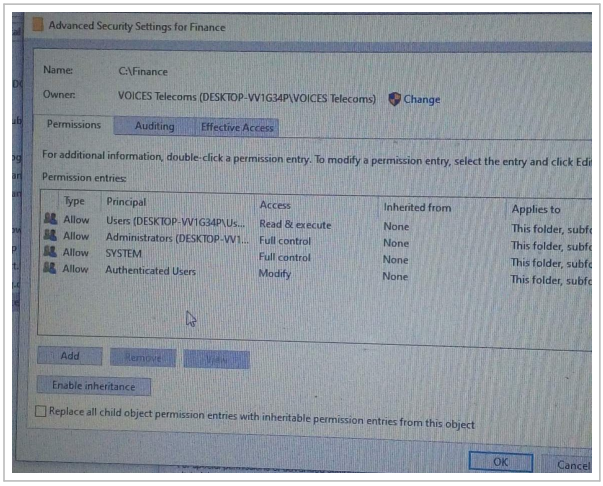
Tunde counts as an authenticated user, so he would have been able to modify files even after I removed the Users group. This entry had to go as well.

Screenshot: Finance Properties, Security tab, with Authenticated Users selected.

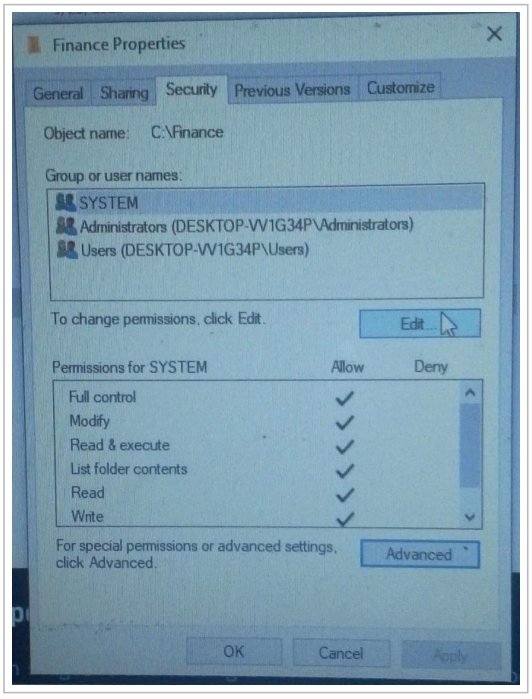
Step 7: Disable inheritance on Finance

I clicked **Advanced** and then **Disable inheritance**, and chose **Convert inherited permissions into explicit permissions on this object**. This keeps the current permissions but stops Finance from receiving future changes from C:.

The window now shows the button **Enable inheritance** (meaning it is currently off) and **Inherited from: None** on every row.



Screenshot: Advanced Security Settings for C:\Finance with inheritance disabled.

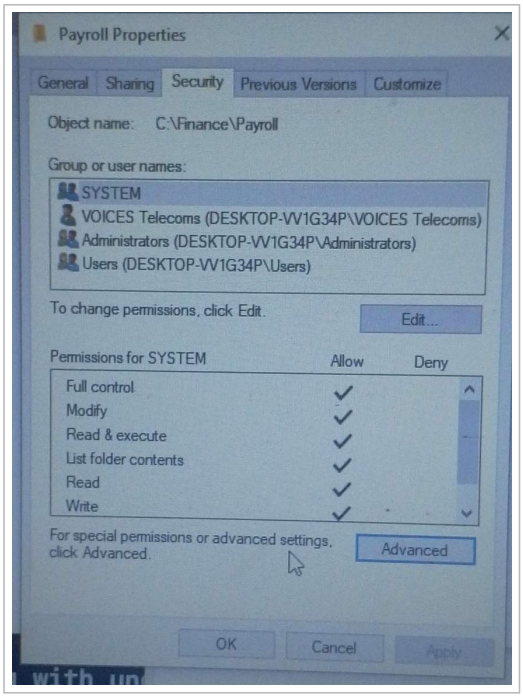


Step 8: Remove Authenticated Users from Finance

With inheritance off, I selected **Authenticated Users** and clicked Remove, then Apply.

Finance now lists only **SYSTEM**, **Administrators** and **Users**, and the ticks are solid because they are set directly on this folder.

Screenshot: Finance Properties after removing Authenticated Users.

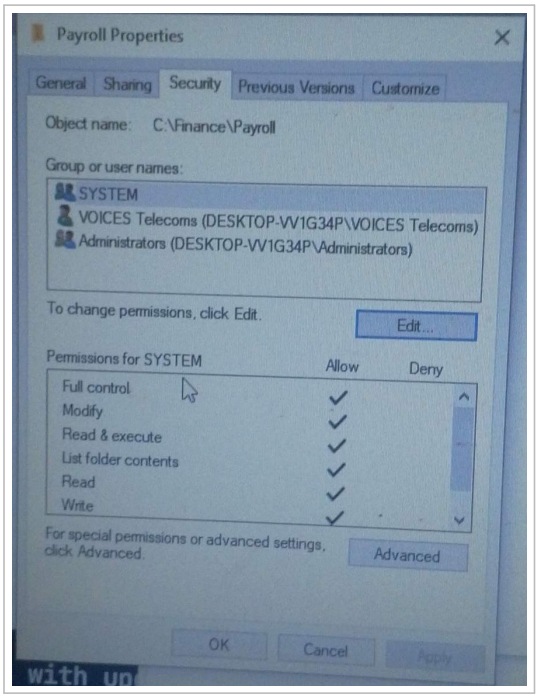


Step 9: Check Payroll again

Back on **Payroll**, the permission ticks were now solid and the list showed SYSTEM, VOICES Telecoms, Administrators, and **Users**.

Because Users still had access here, Tunde could still get into Payroll. I needed to remove Users from this folder.

Screenshot: Payroll Properties before removing Users.



Step 10: Lock Payroll

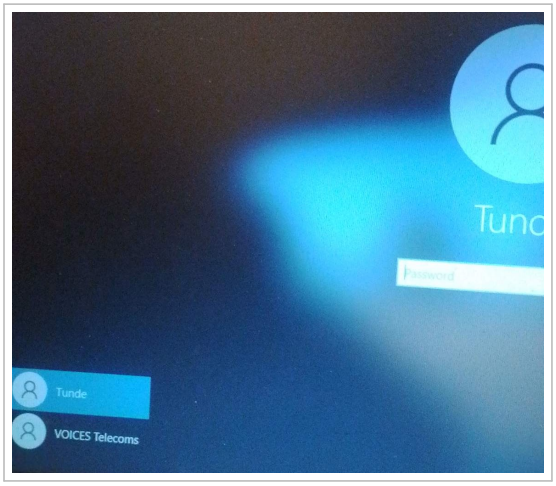
With Payroll's own inheritance already off, I clicked Edit, selected **Users**, and clicked Remove, then Apply and OK.

Payroll now lists only **SYSTEM**, **VOICES Telecoms** and **Administrators**. Tunde is not on this list, directly or through a group, so he has no access.

Screenshot: Payroll Properties after removing Users.

Step 11: Sign in as Tunde

I signed out of my account, and the sign-in screen listed both accounts, **Tunde** and **VOICES Telecoms**. I selected Tunde and entered his password.

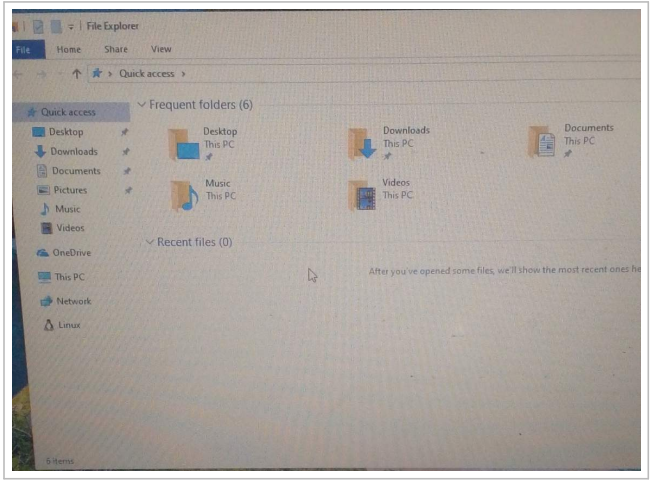


Screenshot: Windows sign-in screen with Tunde selected.

Step 12: Tunde's own profile

On first login Windows built a new, separate profile for Tunde (C:\Users\Tunde). His File Explorer opened with default folders and none of my files, which shows that each account has its own private space.

The C: drive and the Finance folder are shared, so permissions decide what he can open there.

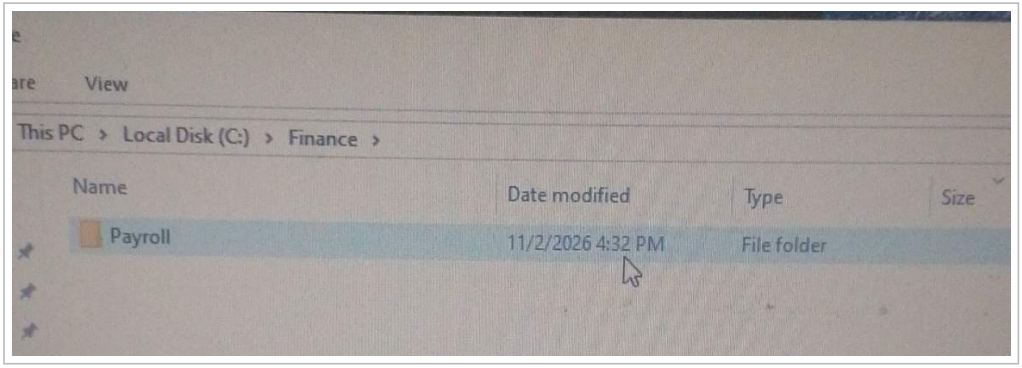


Screenshot: Tunde's File Explorer on first login (Quick access view).

Step 13: Tunde opens Finance

I went to C:\Finance in the address bar. Finance opened because the Users group (which Tunde belongs to) has Read permission, and the list of items inside, **Payroll**, is visible.

Seeing the name is allowed: the Read permission includes List folder contents. Opening it is a separate permission.

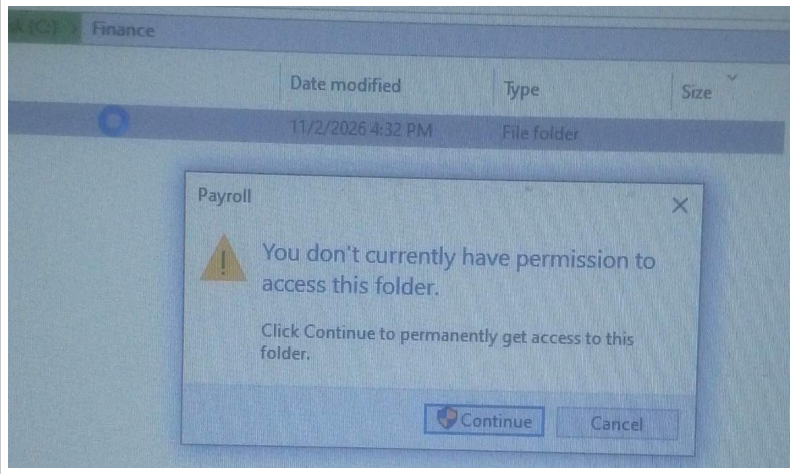


Screenshot: C:\Finance as Tunde, showing the Payroll folder inside it.

Step 14: Tunde tries to open Payroll: access blocked

When Tunde double-clicked Payroll, Windows displayed **"You don't currently have permission to access this folder."** I clicked **Cancel**, not Continue, because Continue prompts for an administrator password and could permanently grant access.

This is the expected result and proves the lab worked.



Screenshot: Access denied message for Payroll, taken on 4 November 2026.

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5. Results

Folder	Who is on the list	Tunde (standard user)
C:\Finance	SYSTEM, Administrators, Users (Read)	Can open and list contents
C:\Finance\Payroll	SYSTEM, VOICES Telecoms, Administrators	Access denied

6. What I learned

- **Permissions are a list of names.** If you are not on the list, you are out.

- **Inheritance saves work but can leak access.** Anything that is allowed higher up (here, Authenticated Users on the C: drive) can reach folders below unless inheritance is cut.

- **Faded tick = inherited, solid tick = set on this folder.**

- **Disabling inheritance is how to give a sensitive folder a stricter list than its parent.** I chose "Convert inherited permissions into explicit permissions" so I did not lock myself out.

- **Never remove SYSTEM or Administrators**, and avoid removing your own account from a folder.

- **Avoid Everyone with Full control.** It is the mistake highlighted in my course diagram.

- **Test as a real second user.** Reading the settings is not enough; the access denied message is the proof.

- **Each account has a private profile; the rest of C: is shared**, so permissions matter on shared locations.

7. Notes and next steps

- The test account **Tunde** still exists on this laptop and uses a simple test password. I should remove it when I no longer need it with `net user Tunde /delete` in an administrator Command Prompt.

- This lab used local accounts on one PC. In a company, permissions go to **groups** (like Finance-Readers and HR) managed in Active Directory, and folders are shared over the network.

- **Next:** install the Windows Server 2022 evaluation in VirtualBox, set up Active Directory, create the Finance-Readers and HR groups, and rebuild the D:\Shares\Finance structure from the course diagram with a network share.